2.1. Data Analysis

1. Identification of Suspicious Behaviors

Based on the available variables (`user_id`, `device_id`, `has_cbk`, CNP environment), a typical exploratory analysis would reveal the following suspicious patterns:

Pattern	Description	Justification
High chargeback rate per device	Devices with a high proportion of `has_cbk` (chargebacks) relative to total transactions.	May indicate the device is being used by a fraudster or has a vulnerability.
Multiple users on the same device	The same `device_id` appearing with several different `user_id` in a short period.	Indicates device sharing or use of emulators to create multiple identities.
High-value transactions with chargeback	Transactions with values well above the user's average that resulted in chargeback.	May indicate card testing or high-value fraudulent purchases.
Irregular time patterns	Spikes in transactions during atypical hours (e.g., late night) or in rapid succession.	Typical behavior of automated attacks.
Transactions without chargeback but with unusual characteristics	e.g., new device_id, high value, no history.	May represent fraud attempts that haven't yet been disputed.

Immediate Actions:

- Prioritize investigation of devices with highest chargeback rates.
- Temporarily block devices showing multiple users in short intervals.
- Establish per-user/device value limits for non-customary transactions.

2. Other Relevant Data Types

For more robust fraud detection, additional data should be incorporated:

- **Geolocation**: IP address, country, city, GPS coordinates (if available). Helps identify transactions from high-risk locations or incompatible with user profile.
- **User History**: Account age, purchase frequency, average amounts, chargeback history. Enables behavioral deviation calculation.
- **Card Data**: BIN (first digits), card type (credit/debit), expiration date, issuing country. Useful for cross-referencing with user information.
- **Transaction Data**: Entry channel (web, mobile), browser type, operating system, plugins. May indicate proxy or anomalous environment usage.
- **Merchant Data**: Product/service category, segment average amount. Allows assessment of transaction compatibility with the business.

- **Attempt History**: Authentication failures (incorrect CVV, wrong password) before successful transaction.
- **User Behavior Data**: Time spent on page, mouse movements, typing patterns (for web applications).

3. Fraud and Chargeback Prevention Measures

Based on the analysis, the following measures are recommended:

- **Dynamic Rules**: Score transactions based on multiple factors (e.g., new device + high value + distant location), with thresholds for approval, rejection, or manual review.
- **Continuous Monitoring**: Monitoring dashboards (like the one provided) that display key metrics in real-time, such as chargeback rate by device, transaction volume by user, and anomaly alerts.
- **Blocklists**: Maintain blacklists of devices, IPs, emails, etc., based on fraud history.
- **Strong Authentication**: Implement 3D Secure (e.g., Verified by Visa, Mastercard SecureCode) for higher-risk transactions, reducing chargeback probability for unauthorized use.
- **Merchant Training**: Guide merchants on security best practices, such as requiring CVV, verifying customer data, and using AVS tools.
- **Post-transaction Analysis**: Perform manual or automated reviews of transactions with high risk scores, even if approved, to detect emerging patterns.

4. Proposed Anti-Fraud Solution

Proposed Solution: Hybrid System (Rules + Machine Learning)

- **Layer 1 – Pre-processing and Enrichment**: Aggregate data from internal sources (user history, device, location) and external sources (fraud databases, suspicious IP lists).
- **Layer 2 – Rules Engine**: Apply a set of heuristic rules (e.g., if `device_id` is new and `amount` > 3x user average → increase score; if `has_cbk` in previous transactions from same device → block).
- **Layer 3 – Machine Learning Models**: Use supervised models (Random Forest, XGBoost) trained on historical transactions and their `has_cbk` outcomes, plus derived features. The model would generate a fraud probability.
- **Layer 4 – Automated Decision**: Combine the rules score and model probability into a decision system (e.g., if score > 0.8 → reject; between 0.5 and 0.8 → manual review; below 0.5 → approve).
- **Layer 5 – Feedback Loop**: Feed the system with results from manual reviews and confirmed chargebacks, periodically retraining models.

Advantages:

- Adaptation to new fraud patterns (via ML).

- Transparency (explainable rules) and speed (ms-level decisions).
- Continuous evolution capability.

Technical Requirements:

- Cloud infrastructure for real-time processing.
- Integration with payment systems and gateways.
- Data science team for model maintenance.

5. Presentation of Results

For the presentation, I would use the **HTML dashboard** (presumably containing interactive visualizations) as the primary communication tool. The dashboard could be complemented with:

- **Distribution Charts**: Histogram of transaction amounts by `has_cbk` to visualize differences between transactions with and without chargebacks.
- **Summary Tables**: List the top 10 devices with highest chargeback rates, users with highest suspicious transaction volumes.
- **Heatmaps**: Correlation between variables (e.g., `device_id` and `user_id`) to detect connections.
- **Time Series**: Transaction volumes and chargebacks over time to identify spikes.
- **Interactive Filters**: Allow analysts to filter by period, device, user, etc.

Additionally, I would prepare a **narrative slide deck** with:

1. **Context** of the challenge and objectives.
2. **Methodology** of analysis (variables considered, statistical methods).
3. **Key Findings** (top 3 suspicious patterns).
4. **Recommendations** (additional data, preventive measures, anti-fraud solution).
5. **Live Dashboard Demonstration**, exploring the charts to validate conclusions.

2.2. Understand the Industry

1. Money Flow, Information Flow, and Roles of Main Players

Money Flow (physical/liquid):

- The cardholder (customer) makes a purchase at a store (online or physical).
- The acquirer receives the transaction and forwards it to the card network (e.g., Visa, Mastercard).
- The card network consults the issuing bank for authorization.

- The issuer authorizes or declines, and funds are transferred from the issuer to the acquirer (after settlement).
- The acquirer deposits the amount into the merchant's account (minus fees).

Information Flow (data):

- The transaction is initiated at the point of sale (e.g., website, app).
- Information (card data, amount, merchant) is sent to the payment gateway, which transmits it to the acquirer.
- The acquirer sends it to the issuer via the card network.
- The issuer returns authorization or decline, and the response is relayed back to the merchant/customer.
- Additionally, risk and historical information is exchanged among players for anti-fraud decisions.

Main Players:

- **Issuer**: Bank that issues the card to the consumer. Responsible for authorization and payment to the acquirer.
- **Acquirer**: Bank or institution that processes payments for the merchant, capturing transactions and receiving funds.
- **Card Network**: Entity connecting issuers and acquirers, establishing rules and fees (e.g., Visa, Mastercard, American Express).
- **Merchant**: Offers goods/services and accepts card payments.
- **Payment Gateway**: Technology provider acting as intermediary between merchant and acquirer, facilitating communication and security (e.g., PayPal, Stripe).
- **Cardholder**: Card owner, who makes the purchase.

2. Differences Between Acquirer, Sub-Acquirer, and Payment Gateway

Player	Description	Key Differences
Acquirer	Financial institution that enrolls the merchant and processes transactions directly with card networks.	Has direct relationship with card networks and assumes settlement risk.
Sub-Acquirer	Company acting as intermediary between merchant and acquirer, typically for smaller merchants.	No direct contract with card networks; outsources processing to a main acquirer.
Payment Gateway	Technology provider connecting merchant to acquirer/sub-acquirer, offering integration, security, and tokenization services.	Does not process financially; it's a software layer that translates and forwards data.

The payment flow changes as follows:

- Merchant → Gateway → Sub-Acquirer → Acquirer → Card Network → Issuer.
- The gateway adds a security layer and facilitates multiple integrations but does not handle financial settlement.

2.3 Chargeback vs. Cancellation

- **Chargeback**: Occurs when the cardholder disputes a transaction with the issuer, who reverses the amount and charges the acquirer. Typically associated with fraud (e.g., unauthorized use) or product/service issues. Chargebacks generate costs and penalties for the merchant and are strong fraud indicators.
- **Cancellation (Refund)**: A request made by the merchant or customer to reverse a transaction before settlement, usually due to purchase remorse or error. It does not involve issuer dispute and does not carry the same risk implications as chargebacks.

Connection to Fraud: In the acquiring world, chargebacks are often the only explicit sign of fraud, indicating that the cardholder does not recognize the purchase. Therefore, monitoring chargeback rates and analyzing transactions that generate them is crucial for detecting fraudulent patterns and preventing future losses.

2.4 What is an Anti-Fraud System and How Acquirers Use It

- **Anti-Fraud System**: A system or set of rules, statistical models, and/or machine learning algorithms that analyze transactions in real-time or batch to identify suspicious behavior and make automated decisions (approve, reject, manual review).
- **How Acquirers Use It**: The acquirer implements an anti-fraud solution to protect merchants and itself from chargeback losses. The tool evaluates variables such as:
    - User/merchant transaction history.
    - Purchase patterns (e.g., amount, frequency, location).
    - Device data (e.g., geolocation, fingerprint).
    - Customer behavior during the purchase journey.

The anti-fraud system can be integrated into the authorization flow, enabling millisecond-level decisions, or operate in post-processing mode for deeper analysis.